

Report 44 - LumpedShaft Model

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Abstract

This report documents the LumpedShaft model family in OpenHPLjl. Catalogue status: **baseline**.

1 Updated Engineering Context

Single-inertia mechanical shaft. It belongs to the Shafts domain.

2 Generalized Concept Model

The model preserves the domain connector interface while allowing internal fidelity to evolve independently.

3 Alternative Models Available in Literature

This family sits on the common fidelity ladder: boundary/algebraic, lumped dynamic, nonlinear physics-based, and distributed/high-order.

4 Governing Mathematics

$$J\dot{\omega} = \tau_m - \tau_e - D\Delta\omega$$

The production model may add states, constitutive relations, lookup maps, or spatial discretization.

5 OpenHPL / Modelica Interpretation

Report 9 maps this family to the closest OpenHPL, Modelica Standard Library, or OpenIPSL analogue.

6 Julia / ModelingToolkit Implementation

Source: `src/Mechanical/Shafts/LumpedShaft.jl`. The file is either an implemented component or an explicit Julia source specification.

7 Component Connections and Assembly

The model connects through the stable physical connector for its domain and should not expose internal fidelity choices to neighbouring components.

8 Numerical Experiment / Validation

Promotion requires an equation-level test, component experiment, limiting-case comparison, structural check, and connected-system test.

9 Expected Outputs

Expected outputs are the physical states, connector variables, and conservation or residual quantities needed to verify this family.

10 Engineering Interpretation

Keep for reduced electromechanical studies.

Conclusion

LumpedShaft now has an explicit source/report pair. Promotion to baseline requires validated equations, parameters, tests, and reference behaviour.